

Understanding at-sea behaviour of terns using hidden Markov models



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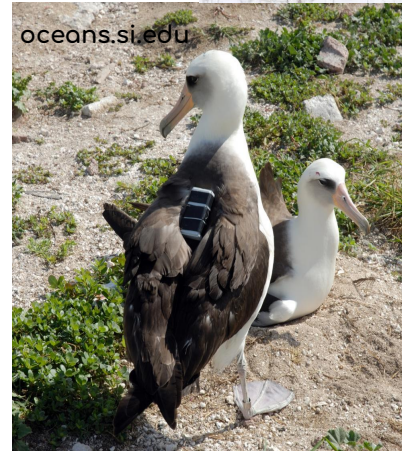
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Movement ecology

- Movement is vital for animal survival
- An active area of research in statistical ecology due to advancement in technology

E.g. GPS devices, biologgers, accelerometers



Movement Data

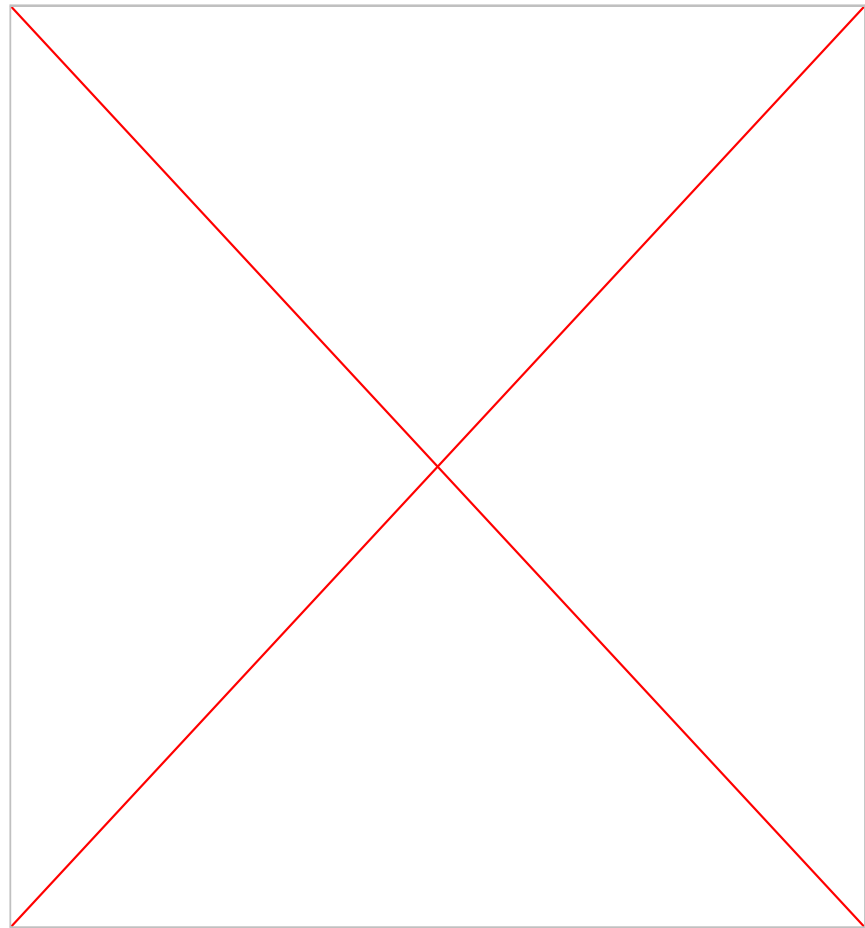
➤ What is interesting about animal movement data?

Directly :

- tell where (location) an animal has been
- reveal how (pattern) an animal moves

Also,

- tell what an animal has been up to - but not directly



Video: www.movebank.org

Movement ecology

- Patterns in animal movement data are influenced by hidden underlying behaviour from time to time
- Identifying behaviours is of great interest in movement ecology
 - conservation management
 - spatial planning
- Underlying behaviours can be inferred from animal movement data
 - using hidden Markov models (HMMs)
 - however, inferred behaviours are rarely validated in practice

Motivation- Seabirds

- Declining at a faster rate than any other type of bird in the UK [1]
- Less protected at sea
- Conservation action
 - Identify foraging areas
 - Establish marine protected areas
- HMMs inferred foraging areas are rarely validated

Kittiwake



Shag



Arctic Skua



Roseate tern



Figure : red listed seabirds in the UK

Goal

- Identify where seabirds forage
 - Fit hidden Markov models (HMMs) to movement data
- Validate HMMs inferred behaviours (foraging and not-foraging)
 - Requires availability of **observed behavioural data**

Rarely available in practice

Study species - Terns

- Forage not too far from the colony
 - Can be followed easily on the boat
- collection of **observed behavioural data** of tern is possible

Rarely available in practice

Arctic tern



Common tern



Roseate tern



Sandwich tern



Tern visual tracking data

- Collected by the Joint Nature Conservation Committee (JNCC) (2009 - 2011)
- Movement data
 - GPS coordinate of the boat used in tracking tern
- Behavioural data
 - behaviour of tracked tern recorded by the observer on the boat
- Data is recorded every second

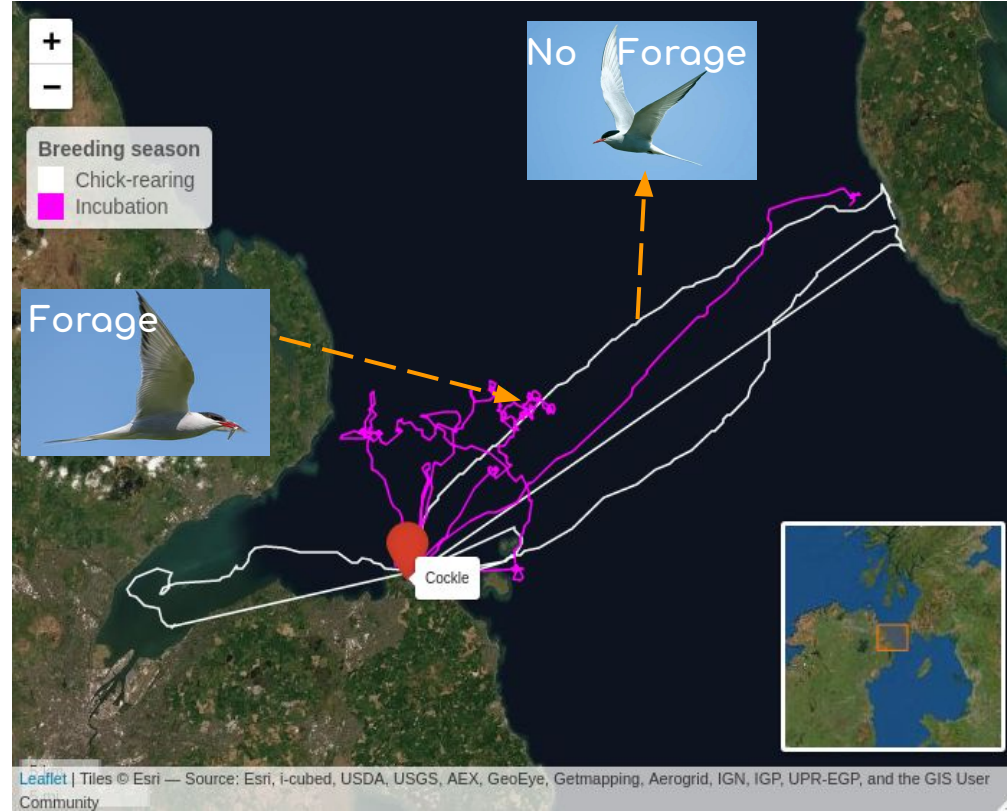
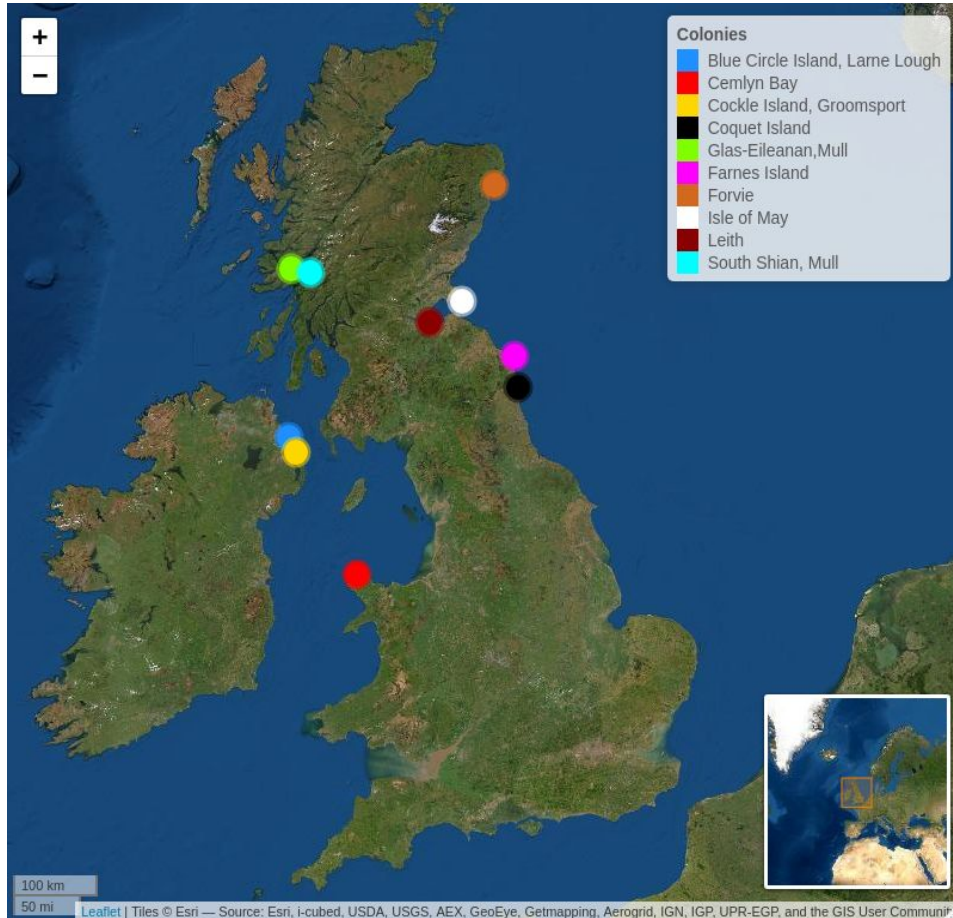


Figure: Visual tracking data of sandwich terns at Cockle Island colony ($4^{\circ}67'N$, $5^{\circ}61'W$), Northern Ireland, 2010

Study Sites



Terns breeding season

Incubation



Usually around
May

Chick-rearing



Usually around
June - July

Assessing Visual tracking data as a good proxy

- Colony: Coquet Island, 2009 data

$$\text{Lat}_{\text{tern}} \approx \arcsin \left(\sin(\text{Lat}_{\text{boat}}) * \cos(\text{distance}/R) + \cos(\text{Lat}_{\text{boat}}) * \sin(\text{distance}/R) * \cos(\text{bearing}) \right),$$

$$\text{Lon}_{\text{tern}} \approx \text{Lon}_{\text{boat}} + \text{atan2}(y, x)$$

where

$$y = \sin(\text{bearing}) * \sin(\text{distance}/R) * \cos(\text{Lat}_{\text{boat}}), \quad \text{and}$$

$$x = \cos(\text{distance}/R) - \sin(\text{Lat}_{\text{boat}}) * \sin(\text{Lat}_{\text{tern}}).$$

$$R, \text{radius of the earth} = 6371000m$$

Assessing Visual tracking data as a good proxy

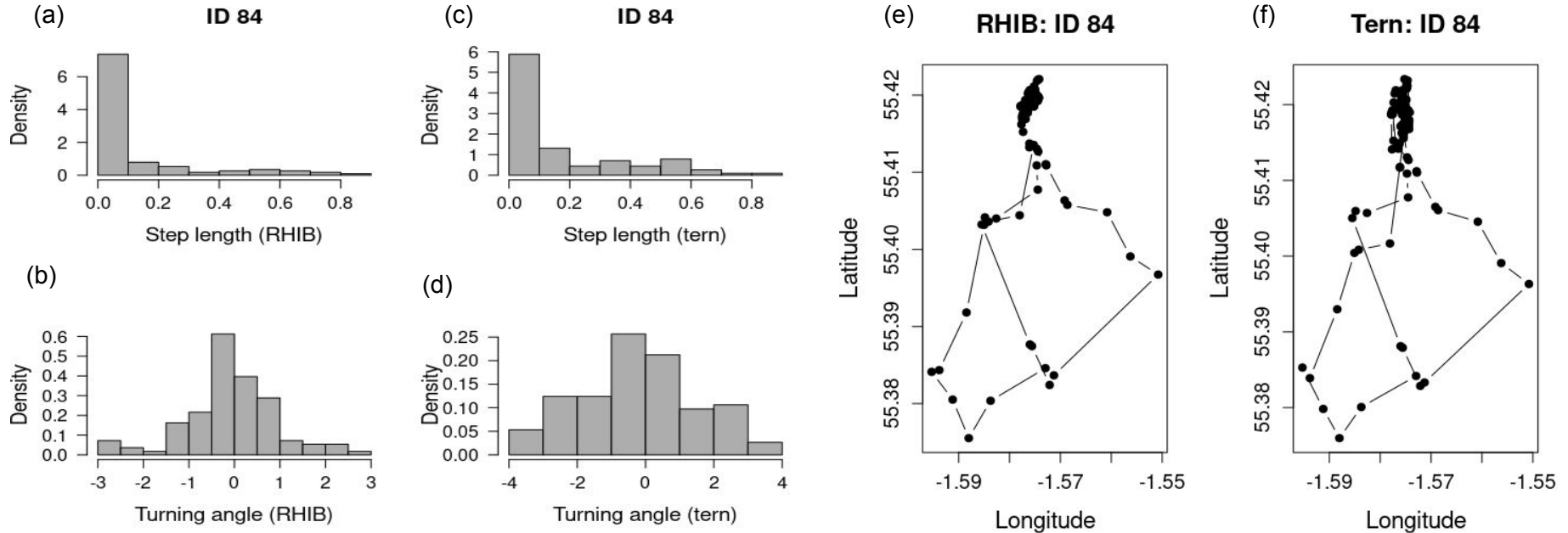


Figure: Histogram of step length and turning angle distribution for visual tracking (a,b) and approximate GPS-tracking (c,d) of roseate tern in Coquet Island colony, 2009. Plot of (e) visual tracking data and (f) approximate GPS-tracking data of a Roseate tern in Coquet Island, 2009.

Hidden Markov model (HMM)

- Time series model with used to infer the hidden from the observed

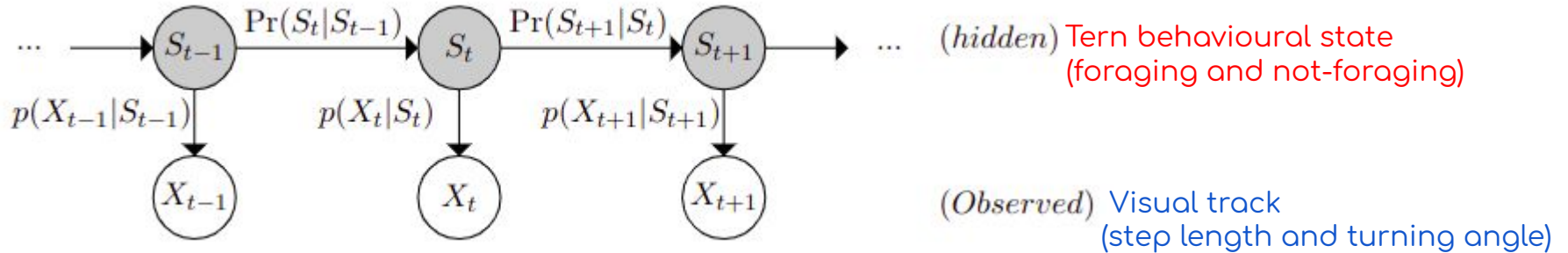


Figure: Representation of an HMM

HMM validation

➤ F1-score :

$$2 \left(\frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

- where precision :

$$\frac{\text{No. of true positive}}{\text{No. of true positive} + \text{No. of false positive}}$$

And recall :

$$\frac{\text{No. of true positive}}{\text{No. of true positive} + \text{No. of false negative}}$$

➤ Log-loss :

$$- [y \log(p) + (1 - y) \log(1 - p)]$$

where

$$y = \begin{cases} 1, & \text{if state is foraging} \\ 0, & \text{if state is not-foraging} \end{cases}$$

And

$$p = \Pr(y = 1) \quad \text{or} \quad \Pr(y = 0)$$

Application

- Complete pool HMM
 - same model parameter across all tracks

Table 1: Summary of visual tracking data of terns across the breeding colonies during the chick-rearing period.

Chick-rearing 2010			
Colony	Tern species	No. of tracks	Total tracking seconds
Isle of May	Arctic	5	3292
South Shian	Common	6	14272
Coquet	Roseate	1	2984
Cockle	Sandwich	5	13014

Table 2: Validation result of HMM fitted to visual tracking data of terns during the chick-rearing breeding season. PPV = positive predictive value (foraging), NPV = negative predictive value (not-foraging), TPR = true positive rate.

Chick-rearing, 2010		Validation metrics				
Colony	Species	PPV	NPV	TPR	F1-score	log-loss
Isle of May	Arctic	0.8641	0.7170	0.7099	0.7794	7.5165
South Shian	Common	0.7026	0.9668	0.8995	0.7889	3.9145
Coquet	Roseate	0.6807	0.9030	0.8372	0.7508	6.6749
Cockle	Sandwich	0.8776	0.9531	0.8602	0.8688	2.3079

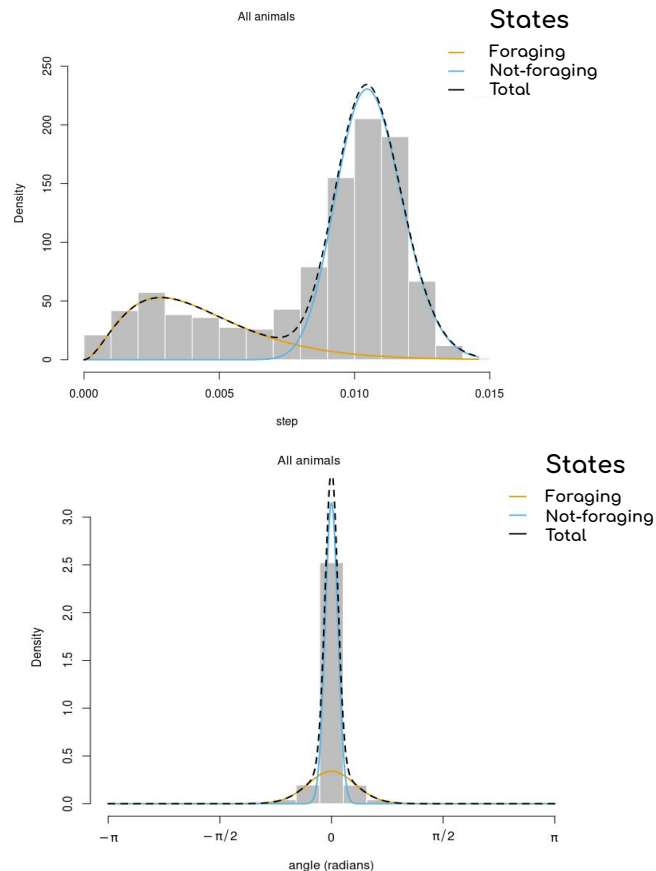


Figure: Histograms of the observed step length (a) and turning angle (b) distributions for visually-tracked common tern in South Shian colony. Lines represent the HMM fitted state-dependent distributions, and are coloured according to decoded behavioural state.

Result

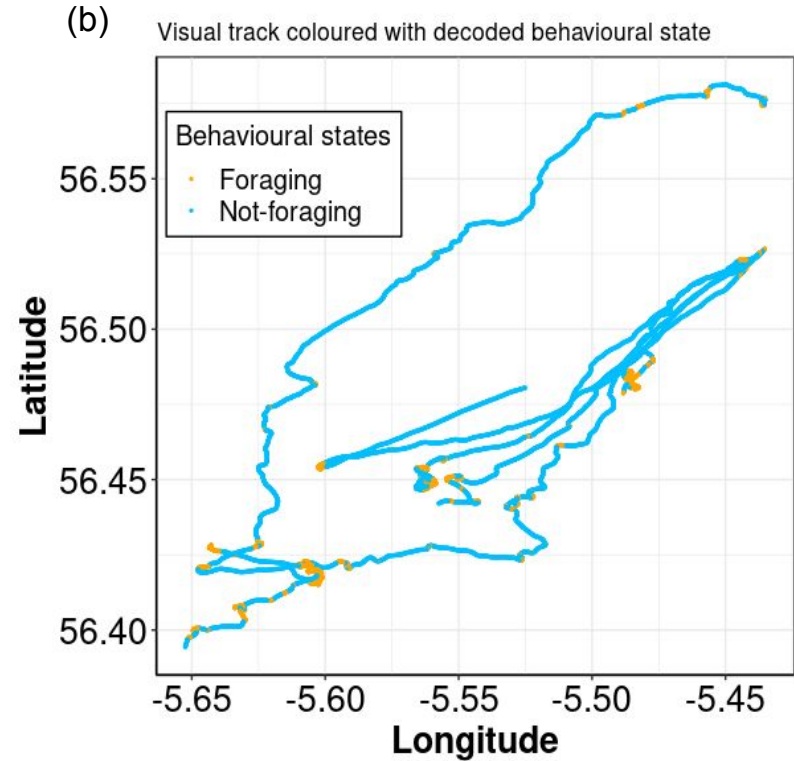
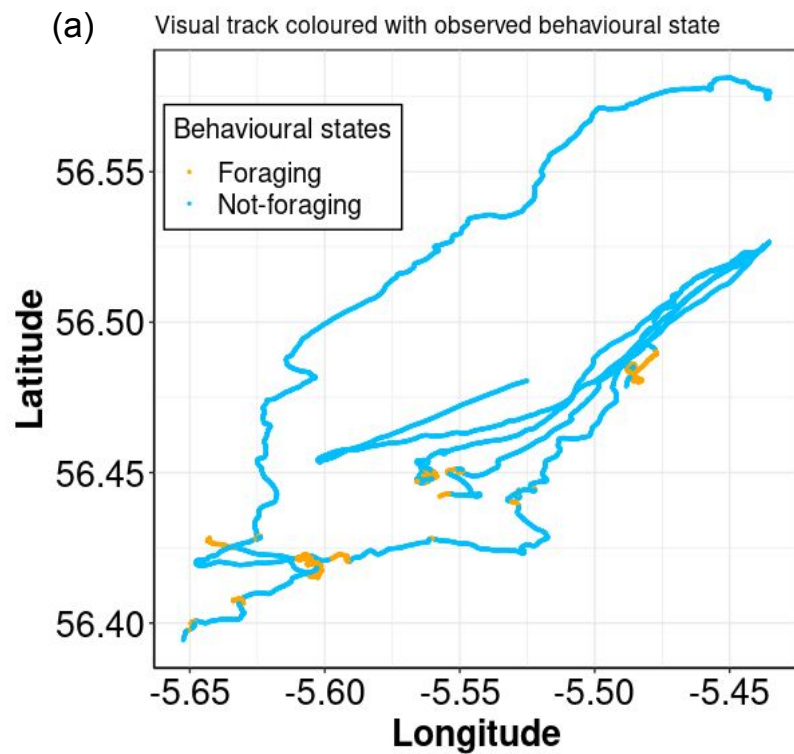


Figure: Visual tracks of common terns in South Shian colony coloured by the (a) observed and (b) decoded behavioural states.

Findings

- Ranking of fitted HMMs based on AIC differs from validation
- However, accuracy across all fitted HMMs are similar
- Treating misclassification in regions of behavioural state transitions improves performance of HMMs
- Visual tracking data is a suitable proxy to GPS-tracks from devices attached to terns

References

1. Lascelles, B.G., Langham, G.M., Ronconi, R.A. and Reid, J.B. (2012), From hotspots to site protection: Identifying Marine Protected Areas for seabirds around the globe. *Biological Conservation*, 156, pp. 5-14.
2. Hidden Markov Models for Time Series; An Introduction Using R
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Thanks



for listening